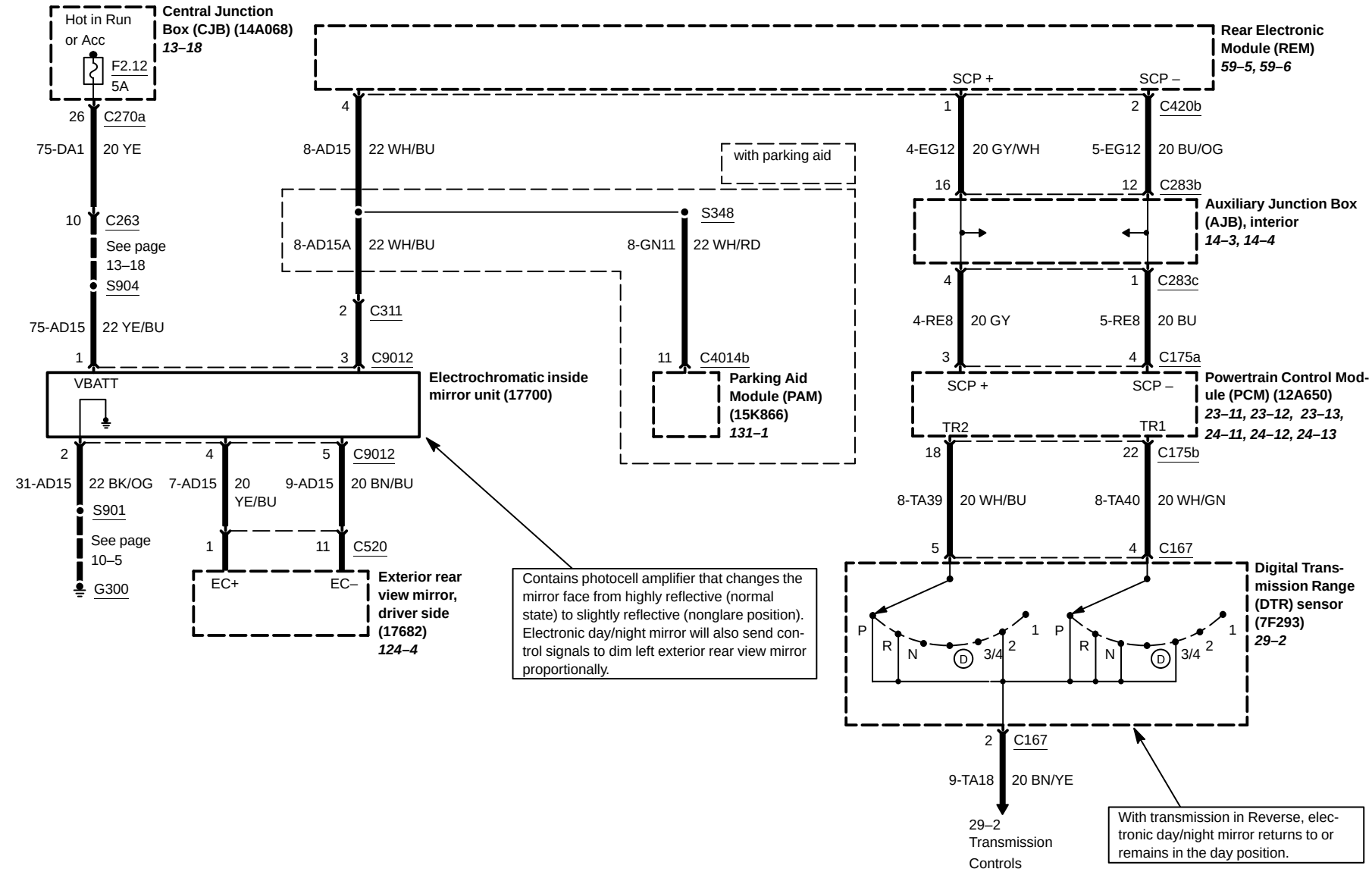without navigation system